

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.6: Surface Area of Solids — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.6: Surface Area of Solids
Driving Question	DRIVING QUESTION: How do we calculate exactly how much material is needed to cover or construct any solid shape — and why does the shape of a container matter even when the volume stays the same?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Nets of Cones: How Much Metal Makes a Bucket?	Three metal bucket models (cylindrical, frustum-shaped, conical) all holding 10 litres; a cardboard net of a cone (sector + circle base) which learners assembled and unrolled; the key observation that the arc length of the sector equals the circumference of the cone's base (arc = $2\pi r$).	The total surface area of a cone = $\pi r l + \pi r^2$, where r is the base radius and l is the slant height; the lateral surface of a cone unfolds into a sector of a circle with radius l ; the slant height $l \neq$ perpendicular height h ; for the KNH paper cup ($r = 3.5$ cm, $l = 8$ cm), TSA = 126.5 cm ² .	The formula $TSA = \pi r l + \pi r^2$ is derived by expressing the sector as a fraction (r/l) of a full circle of radius l , giving lateral area = $\pi r l$, then adding the base area πr^2 ; the shape of a container affects its surface area even when the volume is fixed — this is why the shopkeeper's claim about the frustum bucket needs to be tested, and we will check it fully by computing and comparing all three buckets' surface areas across the unit.
Lesson 2 Surface Area of Pyramids (Square and Rectangular Base)	Learners cut out and folded cardboard nets of square-based and rectangular-based pyramids, labelled all five faces, and measured base dimensions and slant heights. They recorded measurements in a face-by-face area table and verified that the net folded into a closed 3D pyramid.	The total surface area of a square-based pyramid is $TSA = l^2 + 2ls$ (base area plus four triangular faces). For a rectangular-based pyramid, opposite triangular faces form congruent pairs, giving $TSA = lb + ls_1 + bs_2$. Slant height (measured along the triangular face) must be used — not the vertical height of the pyramid. For a square pyramid with $l = 10$ cm and $s = 13$ cm, TSA = 360 cm ² .	The formula is derived by summing the area of the square (or rectangular) base and the areas of all four triangular faces — the same process a jua kali sheet-metal worker in Kamukunji uses when pricing how much flat metal sheet to cut before hammering a pyramid-shaped object. This adds a second data point (after the cone in Lesson 1) to the class comparison of material use across different solid shapes, advancing the investigation of the hardware-shop owner's claim about the frustum.
Lesson 3	Learners peeled an orange and pressed the	The surface area of a sphere equals exactly	Learners applied $SA = 4\pi r^2$ to calculate the

Surface Area of a Sphere	peel flat to cover circles traced from the orange's great circle (equatorial cross-section), observing that the peel filled approximately 4 such circles.	4 times the area of its great circle: $SA = 4\pi r^2$. The total surface area of a closed hemisphere is $3\pi r^2$ (curved surface $2\pi r^2$ plus circular base πr^2).	curved surface area and painting cost of a hemispherical mosque dome in Mombasa, and computed the sheet metal needed for a spherical water tank in Kisumu. They compared the sphere's surface area to the open-topped cylindrical and conical buckets (open cone, Lesson 8 data) for the same 10-litre volume, treating the sphere as a hypothetical closed fourth shape, not one of the shop's three buckets, advancing the driving question about which bucket shape uses the most material.
Lesson 4 Surface Area of a Hemisphere	Learners sliced a spherical model (halved orange or rubber ball) in half and observed that a hemisphere has two distinct surface components: a curved outer surface and a flat circular base. They physically traced the flat face, measured its radius, peeled the curved surface, and examined a printed hemisphere net — confirming the two-component structure directly from a concrete object.	The curved surface area of a hemisphere = $2\pi r^2$ (exactly half the full sphere surface area of $4\pi r^2$). The total surface area of a hemisphere = $2\pi r^2 + \pi r^2 = 3\pi r^2$, where the additional πr^2 accounts for the flat circular base. These formulas were applied to calculate the canvas needed for a hemispherical safari tent at Maasai Mara and the cost in Kenyan Shillings.	At the mid-unit DQB checkpoint, learners updated their predictions about the three Kirinyaga Road buckets using the surface area formulas learned so far (cone, pyramid, sphere, hemisphere). They articulated revised claims about which bucket uses the most metal and generated new specific questions about the frustum-shaped bucket's surface area — setting the inquiry agenda for Lessons 5 and 6.
Lesson 5 Introduction to Frustums: Nets and Vocabulary	Learners physically cut a paper cone along a plane parallel to its base to produce a frustum, measured its dimensions (R , r , h , l), unfolded its curved surface onto squared paper, and observed that the net is an annular sector — the region between two concentric arcs — which is larger and more complex than the full cone net from Lesson 1.	The curved surface area of a frustum of a cone is given by $\pi(R + r)l$, where R is the radius of the larger circular end (the open top of the bucket), r is the radius of the smaller circular end (the base of the bucket), and l is the slant height. The slant height can be calculated from the vertical height and the two radii using the Pythagorean relationship $l = \sqrt{h^2 + (R - r)^2}$. The total surface area of a closed frustum also includes the areas of the two circular ends: $\pi R^2 + \pi r^2$.	The formula $\pi(R + r)l$ arises because the curved surface of a frustum is the difference between the lateral surfaces of two similar cones sharing the same apex: Curved SA = $\pi RL - \pi r(L - l) = \pi l(R + r)$. This connects directly to the Kirinyaga Road phenomenon: the tapered (frustum) bucket has two different-sized circular ends and a sloping curved surface whose area depends on both radii simultaneously, which is why its net looks larger and more complex than the straight-sided cylindrical bucket or the conical bucket of the same volume.
Lesson 6 Total Surface Area of Frustums of Cones — Open and Closed	Learners measured real frustum-shaped containers (tapered buckets, plastic tumblers) brought from home, recording the top radius R , bottom radius r , and perpendicular height h . They compared computed surface areas across groups and, where available, checked against manufacturer label data. They also observed the derived net of a frustum — two concentric annular sectors plus two	The curved surface area of a frustum of a cone is $\pi(R + r)l$, where $l = \sqrt{h^2 + (R - r)^2}$ is the slant height. A closed frustum's total surface area adds both circular faces: TSA = $\pi(R + r)l + \pi R^2 + \pi r^2$. An open frustum (such as a metal bucket with no lid) uses only TSA = $\pi(R + r)l + \pi r^2$ (R = top radius, r = base radius). Whether the frustum uses more sheet metal than a cylinder or cone of equal volume depends on the specific	Because the frustum is formed by removing a small cone from a large one, its lateral surface unfolds into a large annular (ring-shaped) sector rather than a simple rectangle or triangle. The formula $\pi(R + r)l$ captures the fact that both the large and small circular edges contribute to the boundary of this curved surface. Adding the areas of the one or two circular faces to the curved surface area gives the total amount

	circles — which is visibly larger and more complex than the rectangular net of a cylinder of equal height.	dimensions of each shape — the shopkeeper's claim must be checked against the dimensions of the actual buckets (Lesson 8) and is not a universal rule.	of sheet metal needed — which is why a tapered bucket can require a different sheet of metal than a straight-sided cylindrical bucket of identical volume.
Lesson 7 Surface Area of Frustums of Pyramids	Learners handled physical cardboard models and unfolded nets of a pyramidal frustum, counted and labelled all six faces (two square bases and four trapezoidal lateral faces), and measured the bottom side length (b), top side length (a), and slant height (l) of each trapezoidal face.	The total surface area of a closed pyramidal frustum with square bases is $SA = a^2 + b^2 + 2(a+b)l$, where a is the top side length, b is the bottom side length, and l is the slant height of each trapezoidal face. For an open-top frustum, the top base area (a^2) is omitted.	Learners explained why the formula has three distinct parts — two base areas and a combined lateral area — by connecting each algebraic term to a physically observed and measured face on their cardboard net. They applied this explanation to calculate the sheet metal area of a trophy stand pedestal used at a Nakuru County athletics championship, and verified their result by directly measuring the area of their assembled net.
Lesson 8 Consolidation, Applications and Model Revision: Comparing All Three Buckets and Finalising the Surface Area Model	Learners calculated the surface areas of three buckets — cylindrical, conical, and frustum-shaped — all holding exactly 10 litres of water, using dimensions provided on the Bucket Data Sheet. They displayed their full working on sugar-paper sheets and presented findings to the class. Learners also observed, via the DQB review, the full set of question cards built across eight lessons.	Equal volumes do NOT guarantee equal surface areas — the shape of a solid determines how much material is needed to construct it, independently of its capacity. The frustum's formula $\pi(R+r)l + \pi r^2$ accounts for two different circular radii and a slant height, making it more complex than the cone or cylinder formula. Under the specific equal-volume dimensions used in this lesson, the open cone required the greatest surface area of sheet metal. Each of the five solid types studied (cone, pyramid, sphere, hemisphere, frustum) has a unique surface area formula derived from its net.	The shopkeeper's claim that the frustum uses the most metal is not universally correct — it depends on the specific dimensions chosen. For our 10-litre buckets, the open cone used more sheet metal than the frustum. The shape of a container matters for material cost because surface area and volume are related by different geometric formulae: changing the shape while keeping the volume fixed changes the surface area. This is why engineers and manufacturers choose specific solid shapes — to minimise material cost, maximise strength, or meet practical requirements like stackability. In Kenya, frustum-shaped buckets are common despite potentially higher material costs because their tapered shape makes them easier to stack and carry, demonstrating that mathematical optimisation in real life must balance multiple competing factors.

ATTRIBUTION AND LICENSING

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